

# Custom MQL5 Functions

## Overview

This document describes the custom MQL5 functions implemented for advanced trading and management in MetaTrader 5. Each function is explained with its purpose, return values, and example usage.

## Function Descriptions

### 1. `fxIsBarOpen()`

- **Description:** Checks if a new bar (candle) has formed on the chart.
- **Returns:**
  - `true` if a new bar has opened.
  - `false` otherwise.
- **Example Usage:** Executes code only on new bar openings.

### 2. `fxConvertPipToReal(double argPip)`

- **Description:** Converts the specified number of pips into a real price value.
- **Returns:** The converted value based on the decimal precision of the instrument.
- **Example Usage:** Calculating SL, TP, or gap values in real price terms.

### 3. `Ask(string argSymbol=NULL)` and `Bid(string argSymbol=NULL)`

- **Description:**
  - **Ask():** Returns the current Ask price for an instrument.
  - **Bid():** Returns the current Bid price for an instrument.
- **Returns:** Normalized Ask or Bid price.
- **Example Usage:** Used for entry, SL, and TP calculations.

#### 4. **fxStopLoss(int argType, double argOpenPrice, int argSL)**

- **Description:** Calculates the Stop Loss (SL) value based on the order type, entry price, and number of pips.
- **Returns:** Normalized SL price.
- **Example Usage:** Setting the SL level for orders.

#### 5. **fxTakeProfit(int argType, double argOpenPrice, int argTP)**

- **Description:** Calculates the Take Profit (TP) value based on the order type, entry price, and number of pips.
- **Returns:** Normalized TP price.
- **Example Usage:** Setting the TP level for orders.

#### 6. **fxAdjustStopLevel(double BaseLevel, double CheckLevel)**

- **Description:** Checks and adjusts a price level (e.g., SL or TP) to meet the broker's minimum stop level requirements.
- **Returns:** Adjusted price level.
- **Example Usage:** Ensures SL/TP levels conform to broker restrictions.

#### 7. **fxBreakEven(int argMagicNumber, int argBreakEvenPip)**

- **Description:** Moves the SL to the breakeven level when profit reaches a specified number of pips.
- **Returns:**
  - **true** if SL is moved to breakeven.
  - **false** otherwise.
- **Example Usage:** Automatic risk management.

**8. `fxTrailingStop(int argMagicNumber, int argActivationPip, int argStepPip)`**

- **Description:** Implements a trailing stop, dynamically adjusting the SL based on profit.
- **Returns:**
  - `true` if SL is updated.
  - `false` otherwise.
- **Example Usage:** Maximizing profits by following market movements.

**9. `fxGetSize(uint argMode, double argRiskPercent, double argStopLossPoints)`**

- **Description:** Calculates the trade volume (lots) based on specified risk (Fixed Fractional).
- **Returns:** The calculated and normalized volume.
- **Example Usage:** Automatic lot size calculation relative to SL and risk.

**10. `fxOpenOrder(...)` and `fxOpenPosition(...)`**

- **`fxOpenOrder()`:**
  - Sends a pending order (Buy Stop, Sell Stop, Buy Limit, or Sell Limit).
  - **Returns:**
    - \* `true` if the order is successfully sent.
    - \* `false` otherwise.
- **`fxOpenPosition()`:**
  - Opens an immediate Buy or Sell position with optional SL and TP.
  - **Returns:**
    - \* `true` if the position is successfully opened.
    - \* `false` otherwise.

**11. `fxPositionCount(int argMagicNumber)`**

- **Description:** Counts the number of open positions associated with a given Magic Number.
- **Returns:** The number of positions.
- **Example Usage:** Monitoring open positions for an EA.

#### 12. `fxActivePositions(int argType, int argMagicNumber)`

- **Description:** Checks if there are open positions of a specific type and Magic Number.
- **Returns:**
  - `true` if such positions exist.
  - `false` otherwise.
- **Example Usage:** Verifying the status of open positions.

#### 13. `fxPendenOrders(int argType, int argMagicNumber)`

- **Description:** Checks if there are pending orders of a specific type and Magic Number.
- **Returns:**
  - `true` if such orders exist.
  - `false` otherwise.
- **Example Usage:** Monitoring pending orders.

#### 14. `fxPendingDelete(int argType, int argMagicNumber)`

- **Description:** Deletes pending orders of a specific type and Magic Number.
- **Returns:**
  - `true` if at least one order is deleted.
  - `false` otherwise.
- **Example Usage:** Managing pending orders.

#### 15. `fxClosePosition(int argType, int argMagicNumber)`

- **Description:** Closes open positions of a specific type and Magic Number.

- **Returns:**
  - `true` if at least one position is closed.
  - `false` otherwise.
- **Example Usage:** Closing specific open positions.

#### 16. `fxClosePositionByDuration(...)`

- **Description:** Closes open positions exceeding a specified duration (in hours).
- **Returns:**
  - `true` if at least one position is closed.
  - `false` otherwise.
- **Example Usage:** Managing positions based on time.

#### 17. `fxIsTradingDay(...)`

- **Description:** Checks if trading is allowed based on the specified days of the week.
- **Returns:**
  - `true` if trading is allowed.
  - `false` otherwise.
- **Example Usage:** Disabling trading on specific days.

#### 18. `DrawSymbol(...)`

- **Description:** Draws a symbol on the chart at a specified point.
- **Example Usage:** Visualizing indicators or annotations on the chart.